INSTAGRAM DATABASE MANAGEMENT SYSTEM

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A Capstone Project on Database Design and Analysis Using PostgreSQL

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Program:
Data Science and Artificial Intelligence

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Technology Used:
* PostgreSQL
* SQL
* pgAdmin4

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TITLE

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ABSTRACT

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This project focuses on the design and implementation of an Instagram Database
Management System using PostgreSQL.

The objective is to understand how social media platforms manage users, posts,
comments, likes, followers, and hashtags.

The project demonstrates relational database design, ER modeling,normalization, SQL implementation, and analytical query processing.

The system successfully stores,retrieves, and analyzes Instagram
user interactions using PostgreSQL

Introduction

Introduction

Social media platforms have transformed the way people communicate, share information, and interact globally. Millions of users access social networking applications daily to upload content, connect with friends, and engage with communities. Managing this massive amount of data requires efficient database systems.

Instagram is one of the world's largest social media platforms, handling billions of interactions daily. These interactions include user registration, posting images and videos, likes, comments, followers, and hashtags.

The objective of this project is to design and implement an Instagram Database Management System using PostgreSQL. The project focuses on database design principles, normalization, entity relationships, and SQL queries to manage and analyze social media data effectively.

This project demonstrates practical implementation of database management concepts in a real-world application environment.

ABOUT INSTAGRAM

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Instagram is a social networking platform that allows users to share photos, videos, stories, reels, and communicate with others through comments, likes, and direct messages.

Company Overview:

Instagram, founded in 2010 by Kevin Systrom and Mike Krieger, has transformed the way people interact, share, and explore visual content online. Acquired by Facebook, Instagram has become a global social media phenomenon, known for its captivating interface and innovative features. With a focus on visual storytelling and user engagement, Instagram has attracted millions of users worldwide, making it one of the leading platforms in the social networking landscape.

Details

Platform Name : Instagram
Launch Year: 2010
Founders: Kevin Systrom and Mike Krieger
Parent Company : Meta Platforms
Category: Social Media
Headquarters : California, USA
Key Features:User profiles,Photo and video sharing,Stories and reels,Comments,Likes
Followers and following,Hashtags
Social interaction

Problem Statement

Problem Statement

Instagram, a global social media phenomenon, has effectively addressed real-world challenges through its innovative product offerings. With a focus on visual storytelling, Instagram empowers users to authentically share their experiences and passions, bridging the gap between digital interactions and genuine connections. By allowing users to share photos and videos accompanied by captions, locations, and hashtags, Instagram provides a solution to the need for meaningful expression and engagement. This core feature solves the problem of connecting in an increasingly digital world, enabling users to form genuine relationships and engage in conversations that transcend geographical boundaries

Instagram's ingenious engagement features, such as likes, comments, and personalised content recommendations through the "Explore" feature, have revolutionised how users interact with content. By addressing the challenge of content overload, Instagram curates relevant content, helping users discover new accounts, posts, and trends tailored to their interests. Furthermore, Instagram's introduction of hashtags has transformed content discovery by allowing users to categorise their posts with keywords. This innovative approach effectively addresses the challenge of navigating through a sea of content to find relevant information, making it easier for users to connect with content aligned with their interests and engage in conversations that matter to them.

In conclusion, Instagram's product design has successfully tackled real-world problems by creating a platform that nurtures creativity, fosters connections, and offers a space for self-expression. Through its diverse features, Instagram addresses the need for authentic engagement, content curation, and meaningful discovery, shaping the digital landscape and providing practical solutions to the evolving needs of its global user base.

PROJECT OBJECTIVES

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Objectives

The main objectives of this project are:

1. Understand Instagram functionality.
2. Design a relational database model.
3. Create normalized database tables.
4. Build an Entity Relationship Diagram.
5. Implement database schema.
6. Insert sample data.
7. Perform SQL operations.
8. Analyze user interaction data.
9. Apply normalization concepts.
10. Understand real-world database implementation.

FEATURES OF INSTAGRAM

FEATURES OF INSTAGRAM

1. **User Profiles:** Instagram allows users to create personal profiles, offering insights into their lives through features such as usernames, full names, bios, and profile pictures. This creates a personalised online presence that reflects each user's identity.

2. **Posts:** A core feature of Instagram is the ability to share photos and videos as posts. Users can add captions, tag locations, and enhance their content using filters, stickers, and other creative tools.

3. **Interactions**: Engagement lies at the heart of Instagram. Users can express appreciation by liking posts and sharing their thoughts through comments. The "Save" feature enables users to bookmark content for later viewing.

4. **Followers and Following:** The platform fosters connections through the "Follow" functionality. Users can follow other accounts to see their posts in their feed, creating a network of connections. Users can also view who is following them, enhancing transparency.

5. **Explore:** The "Explore" feature propels discovery by suggesting content based on user preferences and interactions. Users can explore posts, videos, and stories from accounts they don't follow, fostering a diverse online experience.

6. **Hashtags:** Instagram pioneered the use of hashtags, which categorise posts and enhance discoverability. Users can add relevant hashtags to their posts, making them enhance discoverability. Users can add relevant hashtags to their posts.

REAL- WORLD CHALLENGES

REAL- WORLD CHALLENGES

- Managing millions of users and user information.
- Handling large amounts of data.
- Database scalability
- Data consistency
- Query optimization
- User engagement analysis
- Data security
- Data collection
- Encourages self-expression, and enhances digital interactions.
-
-

CASE STUDY

CASE STUDY

How Instagram Solves Real World Problems?

Problem 1: Building Social Connections in the Digital World
Real-World Challenge
In today's digital age, people often struggle to maintain meaningful social relationships due to geographical distance, busy lifestyles, and limited face-to-face interaction. Users need a platform where they can communicate, share experiences, and stay connected with family, friends, and communities worldwide.
Instagram's Solution
Instagram addresses this challenge by providing features such as posts, stories, reels, likes, comments, direct messages, and follower relationships. These features allow users to share their daily experiences, engage with others, and maintain strong social connections regardless of location.
Database Perspective
To support this functionality, Instagram uses several interconnected entities:
Users
Posts
Comments
Likes
Followers
Messages
The relationship between these entities enables efficient user interaction and content sharing.

Problem 2: Managing Large Volumes of Content

Real-World Challenge
Millions of photos and videos are uploaded daily, making it difficult for users to discover relevant and interesting content.
Instagram's Solution
Instagram uses hashtags, algorithms, explore pages, and recommendation systems to organize and personalize content for users. This helps users quickly discover content related to their interests.

Database Perspective
The database manages content categorization through:
Hashtag table
PostHashtag relationship table
User preference data
Content metadata
These structures improve search performance and content recommendations.

Problem 3: Supporting Content Creators and Businesses
Real-World Challenge
Many individuals and businesses require a platform to promote products, services, personal brands, and creative work.
Instagram's Solution
Instagram provides business accounts, reels, stories, analytics, advertisements, and influencer marketing features that help creators and businesses grow their audiences.
Database Perspective
The database stores:
User profiles
Business profiles
Posts
Engagement metrics
Followers
Interaction history
These relationships help track performance and user engagement.

Problem 4: User Engagement and Interaction
Real-World Challenge
Social media platforms must maintain user engagement to ensure active participation and platform growth.
Instagram's Solution
Instagram increases engagement through:
Likes
Comments
Stories
Reels
Notifications
Follow suggestions
These features encourage users to spend more time on the platform.
Database Perspective
The database records:
Like activities
Comment history
Follower relationships
User interactions
Content engagement statistics

Problem 5: Content Organization Through Hashtags

Real-World Challenge

Users need an effective method to organize and discover content across millions of posts.

Instagram's Solution

Instagram introduced hashtags to categorize content and improve discoverability.

Database Perspective

The following tables support this feature:

Hashtag

Post

PostHashtag (Many-to-Many relationship)

This structure allows a single post to contain multiple hashtags and a hashtag to be associated with multiple posts.

Conclusion of Case Study

Instagram successfully addresses several real-world challenges by providing an efficient social networking ecosystem. The database design plays a critical role in managing users, posts, interactions, relationships, and content categorization. The relational database model ensures data integrity, reduces redundancy through normalization, and supports scalable social media operations

DATABASE DESIGN METHODOLOGY

DATABASE DESIGN METHODOLOGY

Requirement Analysis

Understanding Instagram features and requirements.

Entity Identification

The following entities were identified:
Users
Posts
Comments
Likes
Followers
Hashtags
PostHashtag

Relationship Identification

Relationships between entities were established.

Schema Design

Tables and constraints were created.
Primary key is common constraints to each table.
Foregin Key is used for connect to each table.

Normalization

Normalization techniques were applied. Created many and Constraints.

ENTITY IDENTIFICATION

ENTITY IDENTIFICATION

Entity		Description
Users	-	Stores user information
Posts	-	Stores uploaded posts
Comments	-	Stores user comments
Likes	-	Stores likes
Followers	-	Stores follower information
Hashtag	-	Stores hashtags
PostHashtag	-	Connects posts and hashtags

RELATIONSHIP IDENTIFICATION

Relationships

One User creates many Posts.
One User writes many Comments.
One Post contains many Comments.
One Post receives many Likes.
One User has many Followers.
One Post has many Hashtags

DATA BASE SCHEMA

DATABASE

```
CREATE DATABASE instagram_db;

CREATE TABLE Users(
User_Id SERIAL PRIMARY KEY ,
Username VARCHAR(50) NOT NULL,
Email VARCHAR(200) UNIQUE NOT NULL,
Full_Name VARCHAR(100) NOT NULL,
Bio TEXT,
Registration_Date TIMESTAP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE Post(
Post_Id SERIAL PRIMARY KEY,
User_Id INT,
Caption TEXT,
Image_Url VARCHAR(255) NOT NULL,
Location VARCHAR(200),
Post_Date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
Foreign KEY(User_Id)
REFERENCES  Users(User_Id)
);

CREATE TABLE comment(
comment_Id SERIAL PRIMARY KEY,

CREATE TABLE Likes(
like_Id SERIAL PRIMARY KEY,
Post_Id INT,
User_Id INT,
Like_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY(Post_Id) REFERENCES Post(Post_Id),
FOREIGN KEY(User_Id) REFERENCES Users(User_Id)
);

CREATE TABLE Follower(
Folower_Id SERIAL PRIMARY KEY,
FoLLowing_User_Id INT NOT NULL,
Followers_Use
);
```

DATA INSERTION

Data Insertion

```sql
-- Insert the Data
INSERT INTO Users (Username, Email, Full_Name, Bio)
VALUES
('kareena','kareena@gmail.com','Kareena Gola','Data Science Student'),
('rahul','rahul@gmail.com','Rahul Sharma','Traveler'),
('priya','priya@gmail.com','Priya Singh','Photographer'),
('aman','aman@gmail.com','Aman Verma','Engineer'),
('neha','neha@gmail.com','Neha Gupta','Food Blogger'),
('rohit','rohit@gmail.com','Rohit Kumar','Fitness Coach'),
('simran','simran@gmail.com','Simran Kaur','Fashion Designer'),
('arjun','arjun@gmail.com','Arjun Singh','Musician'),
('pooja','pooja@gmail.com','Pooja Sharma','Teacher'),
('vikas','vikas@gmail.com','Vikas Jain','Developer');

INSERT INTO Post
(User_Id, Caption, Image_Url, Location)
VALUES
(1,'Trip to Manali','img1.jpg','Manali'),
(2,'Morning Coffee','img2.jpg','Delhi'),
(3,'Beautiful Sunset','img3.jpg','Goa'),
(4,'Coding Session','img4.jpg','Bangalore'),
(5,'Delicious Food','img5.jpg','Mumbai'),
(6,'Workout Time','img6.jpg','Delhi'),
(7,'New Dress','img7.jpg','Chandigarh'),
(8,'Music Practice','img8.jpg','Punjab'),
(9,'Teaching Students','img9.jpg','Jaipur'),
(10,'Learning SQL','img10.jpg','Noida');
```

```sql
INSERT INTO Followers
(Follower_User_Id, Following_User_Id)
VALUES
(1,2),
(1,3),
(2,1),
(2,5),
(3,1),
(3,4),
(4,5),
(5,6),
(6,7),
(7,8),
(8,9),
(9,10),
(10,1),
(4,1),
(5,1),
(6,1),
(7,1),
(8,1),
(9,1),
(10,2);
```

Data Insertion

```
INSERT INTO Comment (Post_Id, User_Id, Text)
VALUES
(1,2,'Amazing picture!'),
(1,3,'Beautiful view'),
(2,1,'Nice post'),
(2,4,'Great shot'),
(3,5,'Wonderful'),
(3,6,'Awesome'),
(4,7,'Keep it up'),
(4,8,'Very good'),
(5,9,'Looks delicious'),
(5,10,'Yummy'),
(6,1,'Excellent'),
(6,2,'Super'),
(7,3,'Fantastic'),
(7,4,'Lovely'),
(8,5,'Amazing music'),
(8,6,'Great performance'),
(9,7,'Very inspiring'),
(9,8,'Nice work'),
(10,9,'Helpful'),
(10,10,'Interesting'),
(1,4,'Beautiful'),
(2,5,'Wonderful'),
(3,7,'Nice'),
(4,9,'Excellent'),
(5,1,'Great'),
(6,3,'Fantastic'),
(7,5,'Lovely'),
(8,7,'Amazing'),
(9,2,'Good job'),
(10,4,'Keep learning');
```

Data Insertion

```
INSERT INTO Likes (Post_Id, User_Id)
VALUES
(1,2),(1,3),(1,4),(1,5),(1,6),
(2,1),(2,3),(2,4),(2,5),(2,6),
(3,1),(3,2),(3,4),(3,5),(3,6),
(4,1),(4,2),(4,3),(4,5),(4,6),
(5,1),(5,2),(5,3),(5,4),(5,6),
(6,1),(6,2),(6,3),(6,4),(6,5),
(7,1),(7,2),(7,3),(7,4),(7,5),
(8,2),(8,3),(8,4),(8,5),(8,6),
(9,1),(9,2),(9,3),(9,4),(9,5),
(10,1),(10,2),(10,3),(10,4),(10,5);
INSERT INTO PostHashtag (Post_Id, Hashtag_Id)
VALUES
(1,1),
(1,3),
(2,2),
(2,5),
(3,3),
(3,8),
(4,5),
(4,10),
(5,2),
(5,6),
(6,4),
(6,9),
(7,6),
(7,8),
(8,7),
(8,3),
(9,10),
(9,1),
(10,5),
(10,10);
```

Data Insertion

```sql
INSERT INTO Hashtag (Tag)
VALUES
('#travel'),
('#nature'),
('#food'),
('#fitness'),
('#technology'),
('#coding'),
('#datascience'),
('#ai'),
('#machinelearning'),
('#photography'),
('#fashion'),
('#music'),
('#sports'),
('#education'),
('#motivation'),
('#lifestyle'),
('#health'),
('#adventure'),
('#business'),
('#instagram');
```

DATABASE TABLES

DATABASE TABLES

TABLE NAME	PURPOSE
Users	Users details
Post	Post information
Comment	Comment
Likes	likes tells
Followers	Follower and following
Hashtag	About hashtag used users
Posthashtaag	Post hashtag

SQL QUERY

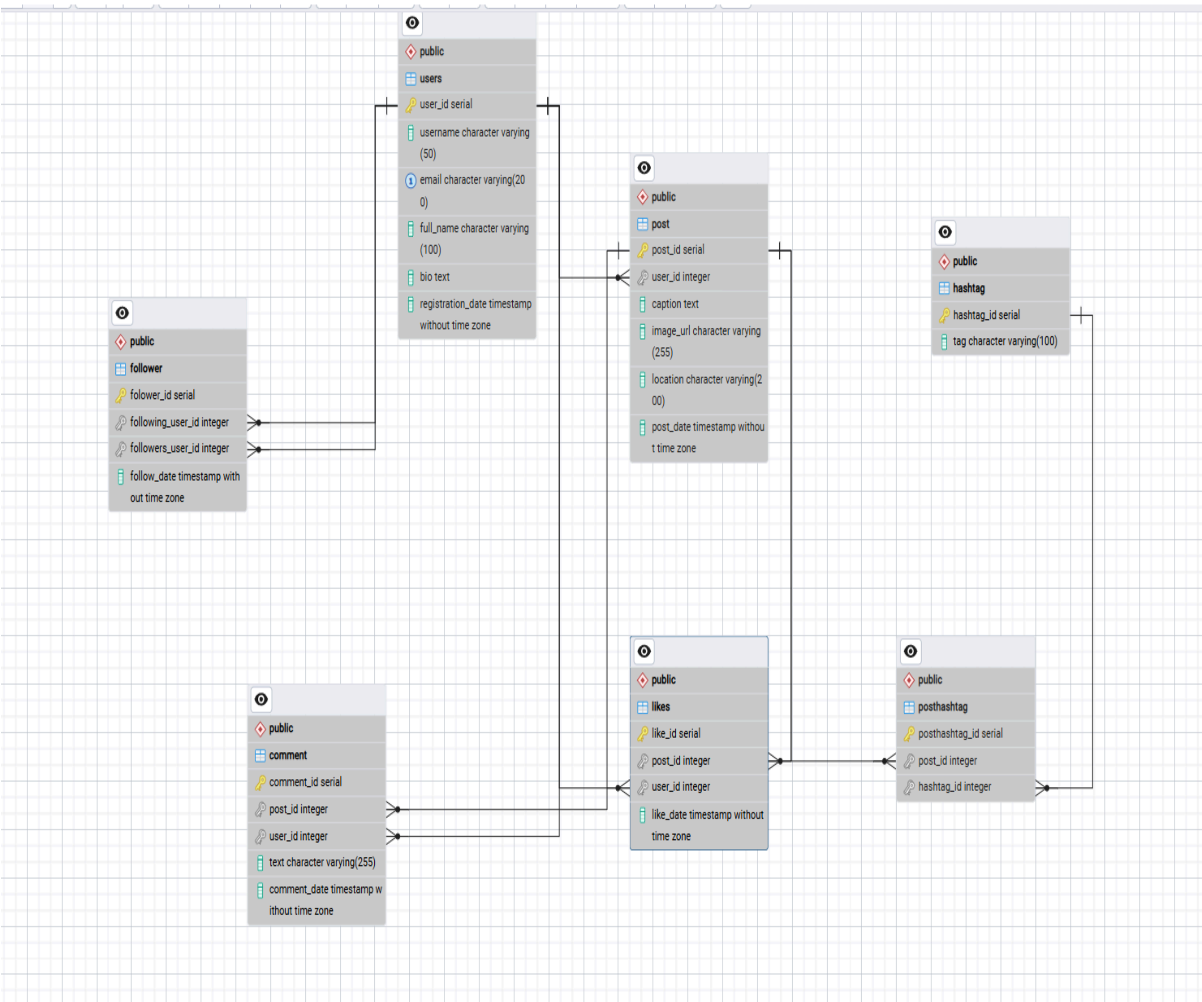
SQL Queries And Analysis

```sql
## QUERY
--VIEW OF TABLE

SELECT * FROM users;
SELECT * FROM Post;
SELECT * FROM Comment;
SELECT * FROM follower;
SELECT * FROM likes;
SELECT * FROM hashtag;
SELECT * FROM posthashtag;

## BASIC QUESTION
-- SHow all post and username
SELECT u.

-- Show all hashtag used posts

-- How many users are registered?
SELECT  COUNT(*) AS total_users FROM users;
-- How many Posts are uploaded?
SELECT COUNT(*) AS total_post FROM post;
-- How many comments exit?
SELECT COUNT(*) AS total_comeent FROM comment;
```

```sql
-- Which users created the most posts?/most_active
SELECT user_id ,COUNT(*) AS total_post FROM post GROUP BY user_id ORDER BY total_post DESC LIMIT 1 ;
-- Which Users has most foloowers?/highest_followers
SELECT followers_user_id ,COUNT(*) AS total_follower FROM follower GROUP BY followers_user_id  ORDER BY total_follower DESC LIMIT 1;
-- find Top 3 post most liked?
SELECT post_id ,COUNT(*) AS total_likes FROM likes GROUP BY post_id ORDER BY total_likes DESC LIMIT 3;
-- which tag most used?
SELECT t.tag,COUNT(p.posthashtag_id) AS most_used_hashtag FROM hashtag t JOIN posthashtag p ON t.hashtag_id=p.hashtag_id
GROUP BY t.tag ORDER BY most_used_hashtag DESC;
-- Which users most comments?/ - more_active
SELECT user_id ,COUNT(*) AS total_comment FROM comment  GROUP BY user_id ORDER BY total_comment DESC LIMIT 1;
-- Find users who never created most a post?
SELECT u.* FROM users u LEFT JOIN post p ON u.user_id =p.user_id WHERE p.post_id IS NULL;
-- Find the post without comment?
SELECT p.post_id FROM post p LEFT JOIN comment c ON p.post_id=c.post_id WHERE c.comment_id IS NULL;
-- Find post without hashatag?
SELECT p.post_id FROM post p LEFT JOIN  posthashtag e ON p.post_id=e.post_id WHERE e.hashtag_id IS NULL;
-- Which post have no likes?
SELECT p.post_id FROM post p LEFT JOIN likes  l ON p.post_id=l.post_id WHERE l.like_id IS NULL;
-- Which users follow the most people?
SELECT following_user_id,COUNT(*) AS most_people FROM follower GROUP BY following_user_id ORDER BY most_people DESC  LIMIT 1;

-- What is the average number of likes per post?
SELECT ROUND(AVG(total_likes),2) FROM (SELECT COUNT(*) AS total_likes FROM likes GROUP BY post_id);
-- What is the average number of comment per post?
SELECT ROUND(AVG(total_comment),2) FROM(SELECT COUNT(*) AS total_comment FROM comment GROUP BY post_id);
-- Which users has the highest engagement?
SELECT user_id ,COUNT(*) AS engagement FROM
(SELECT user_id FROM post UNION ALL SELECT user_id FROM comment UNION ALL SELECT user_id FROM likes ) t GROUP BY user_id
ORDER BY engagement DESC;
```

ER DIAGRAM

ER DIAGRAM



public
users
user_id serial
username character varying (50)
email character varying(200)
full_name character varying (100)
bio text
registration_date timestamp without time zone
public
post
post_id serial
user_id integer
caption text
image_url character varying (255)
location character varying(200)
post_date timestamp without time zone
public
hashtag
hashtag_id serial
tag character varying(100)
public
follower
folower_id serial
following_user_id integer
followers_user_id integer
follow_date timestamp without time zone
public
comment
comment_id serial
post_id integer
user_id integer
text character varying(255)
comment_date timestamp without time zone
public
likes
like_id serial
post_id integer
user_id integer
like_date timestamp without time zone
public
posthashtag
posthashtag_id serial
post_id integer
hashtag_id integer

FINDINGS

FINDINGS

1. User engagement can be measured.
2. Popular posts can be identified.
3. Hashtag performance can be analyzed
4. User interaction pattern can be studied.
5. Database relationships improve query efficiency.
6. Follower and following the tell the users which has crated the more have follower.

CONCLUSION

CONCLUSION

Conclusion

The Instagram Database Management System project successfully demonstrates the practical implementation of relational database concepts using PostgreSQL. The project helped in understanding database design, normalization, ER modeling, SQL queries, and data analysis techniques used in real-world applications.

In this case study, we delved into the design of Instagram's schema and Entity-Relationship diagram. Instagram has revolutionised the way people share and engage with visual content, fostering connections and creative expression. The platform's intricate data model, consisting of entities like users, posts, comments, likes, followers, hashtags, and associations, forms the foundation for its seamless functionality. By understanding this schema, we gain insight into how Instagram effectively manages the complexities of user interactions and content sharing, contributing to its widespread popularity and continued growth in the world of social media.